Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

ITEP 414

System Administration and Maintenance

Week 3 Portfolio Project

Enterprise Server Deployment

and

Operating System Installation

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COLLEGE OF COMPUTER STUDIES

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Professional Installation Guide of Ubuntu Server using Oracle VM VirtualBox

1. Introduction

This professional installation manual for Ubuntu Server Installation provides the procedures for deploying and verifying a clean Ubuntu Server virtual machine for our Week 3 Portfolio Project. The installation follows the given laboratory requirements and uses Oracle VirtualBox as the virtualization platform.

The manual covers the preparation of the virtual machine, Ubuntu Server installation, network configuration, hostname assignment, administrative user creation, disk partitioning, OpenSSH installation, and server verification. It is designed to provide a clear step-by-step procedure that another system administrator can follow without requiring additional instructions.

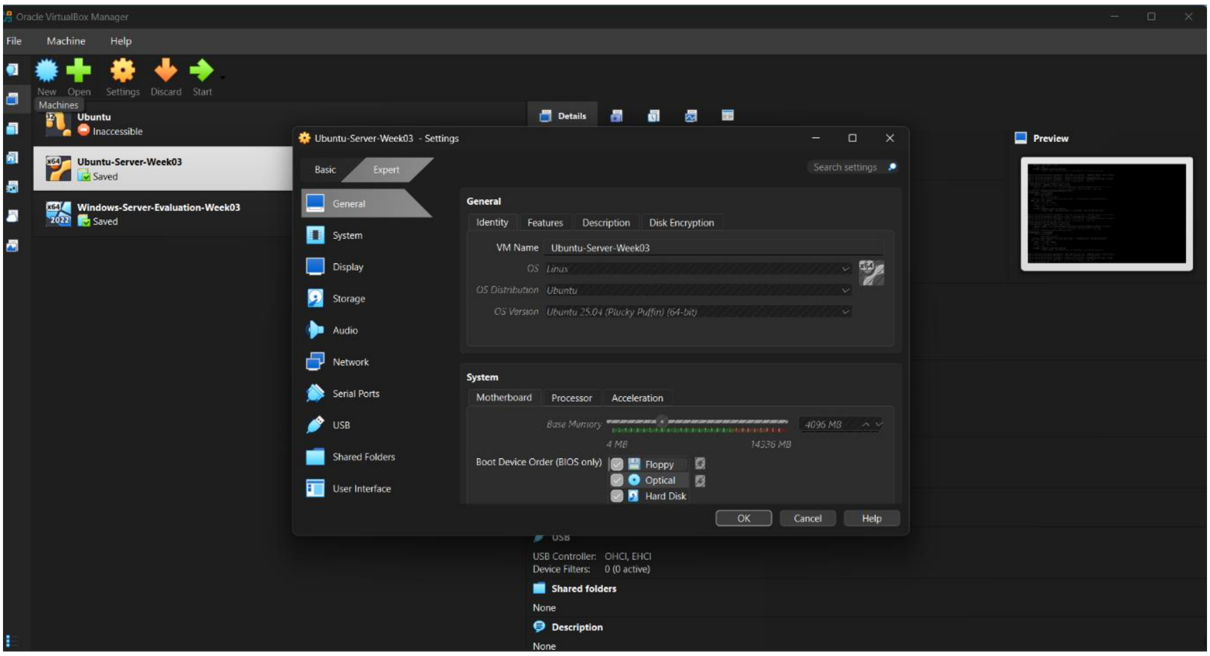
The installation uses the recommended minimum virtual machine specifications provided in the laboratory activity, including 4 GB of RAM, 2 virtual processors, 40 GB of virtual storage, NAT networking, and an Ubuntu Server ISO installation image.

2. Ubuntu Server Installation Guide

This section provides the step-by-step procedure for deploying Ubuntu Server in a virtual machine based on the required laboratory specifications. It covers the virtual machine setup, Ubuntu Server installation, network and hostname configuration, creation of a non-root administrative user, disk partitioning, OpenSSH installation, and completion of the installation. Screenshots are included to document each major installation step.

2.1 Virtual Machine Specifications

Component	Requirement
VM Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB
Disk Type	VDI/VMDK
Network	NAT
Optical Drive	Ubuntu Server ISO



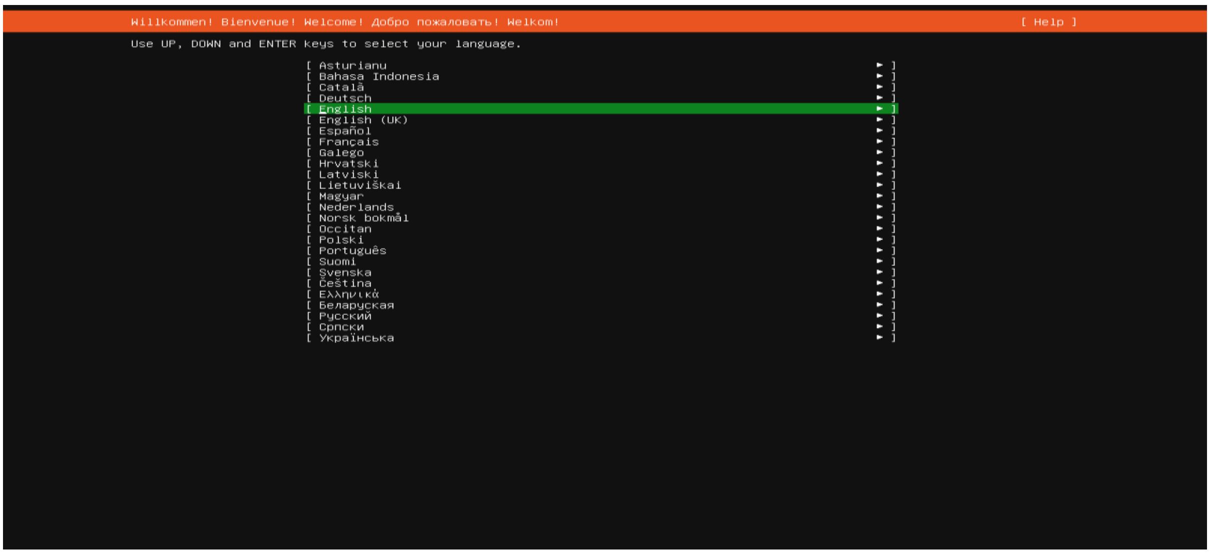
2.2 Installation Tasks

The following procedures describe the required Ubuntu Server installation tasks.

2.2.1 Language

Procedure

1. Start the Ubuntu-Server-Week03 virtual machine.
2. Wait for the Ubuntu Server installer to load.
3. Select: **English**
4. Continue to the next step.

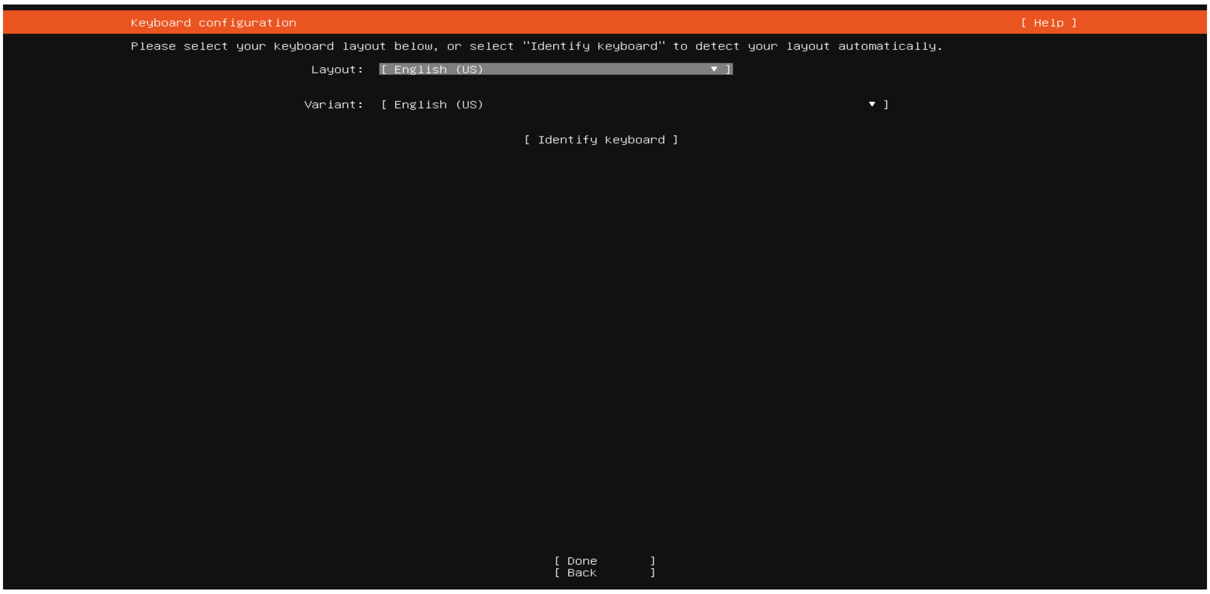


2.2.2 Keyboard Layout

Select the appropriate keyboard layout.

For example: ***English (US)***

Then continue.



2.2.3 Network Configuration

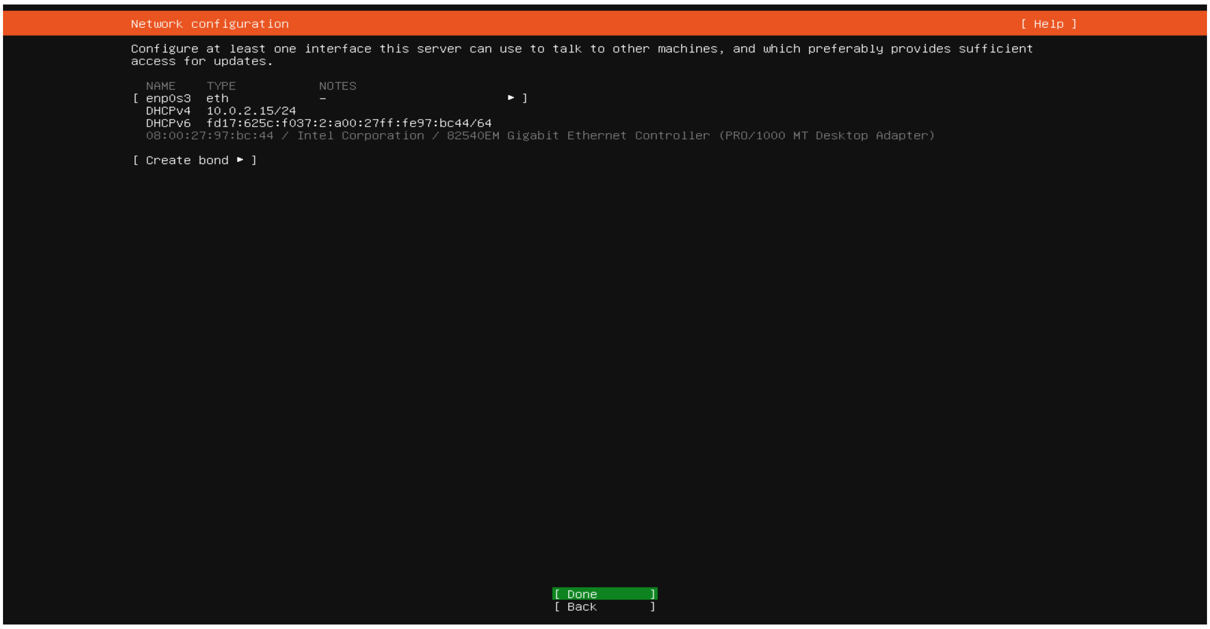
Ubuntu Server should detect the virtual network adapter.

Allow Ubuntu to obtain an IP address automatically.

Important

Record the assigned IP address.

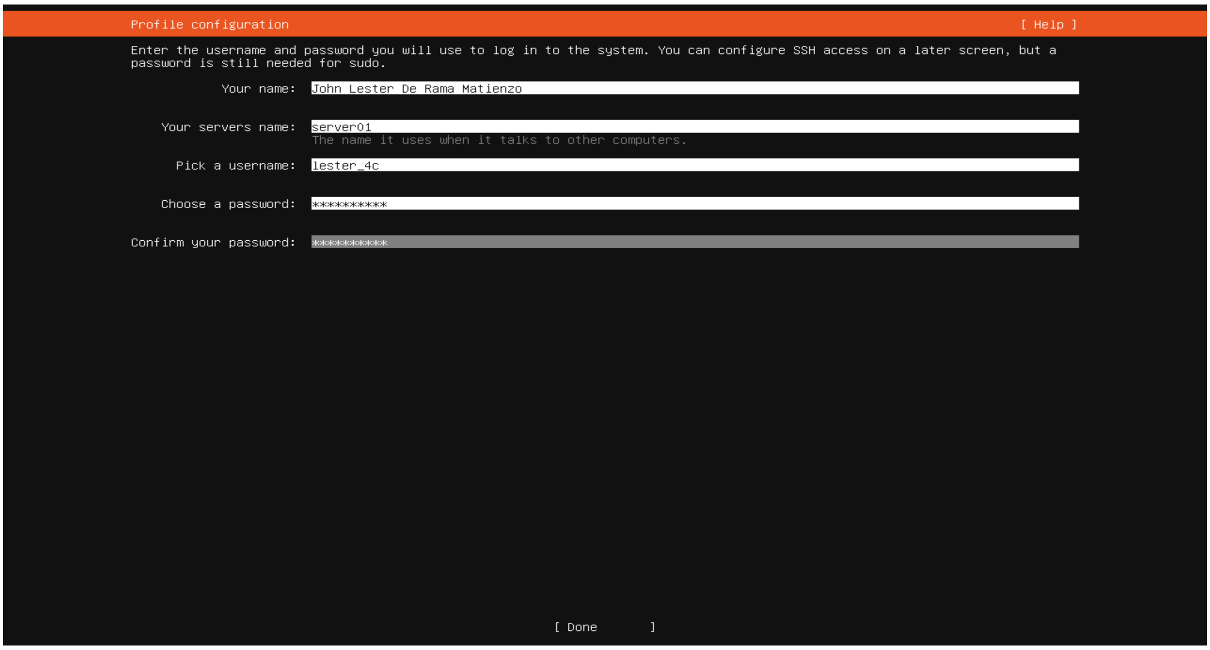
But use the actual IP address assigned to your VM.



2.2.4 Hostname

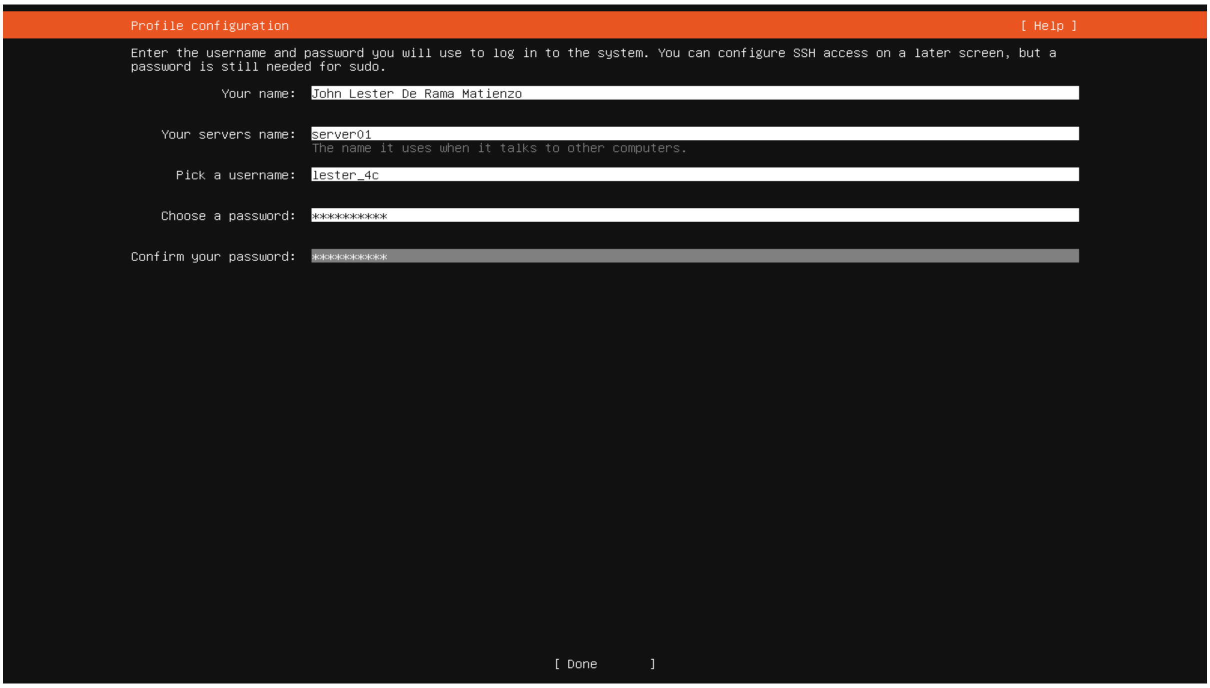
When Ubuntu asks for the hostname, enter exactly what on the instruction

Do not use ubuntu-server if your laboratory specifically requires server01.



2.2.5 User Account

Create a **non-root administrative user**.



2.2.6 Disk Partition

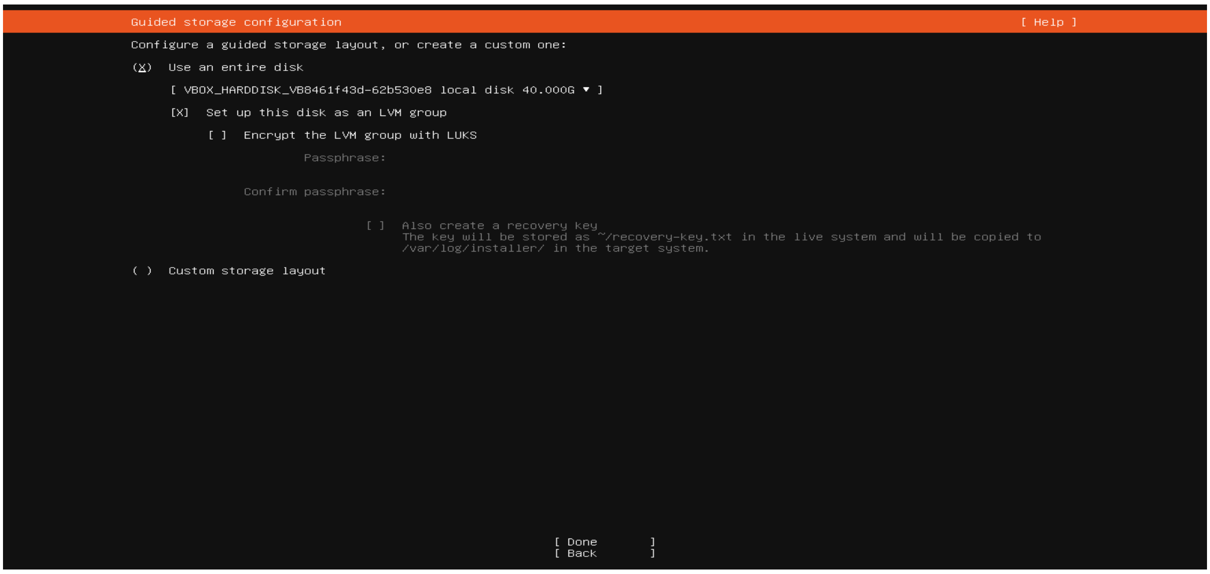
Select:

Guided – Use Entire Disk

Select the 40 GB virtual disk.

Before confirming, document:

- Partition scheme
- File system
- Disk size

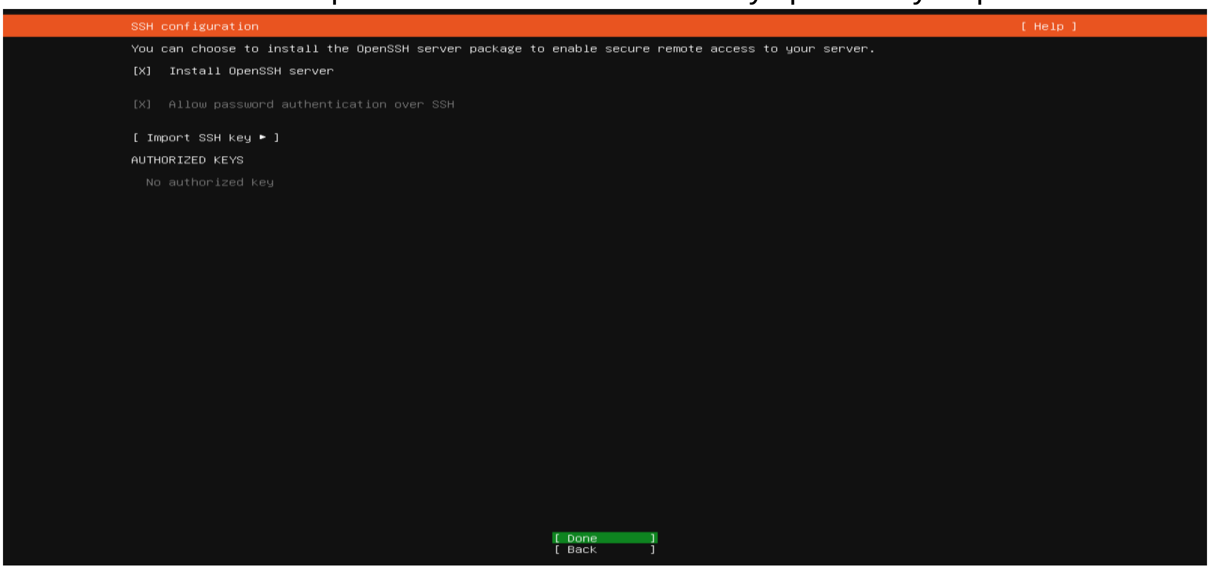


2.2.7 SSH Server

When you reach the SSH configuration screen:

Select: ✓ Install OpenSSH Server

This is important because the laboratory specifically requires SSH.



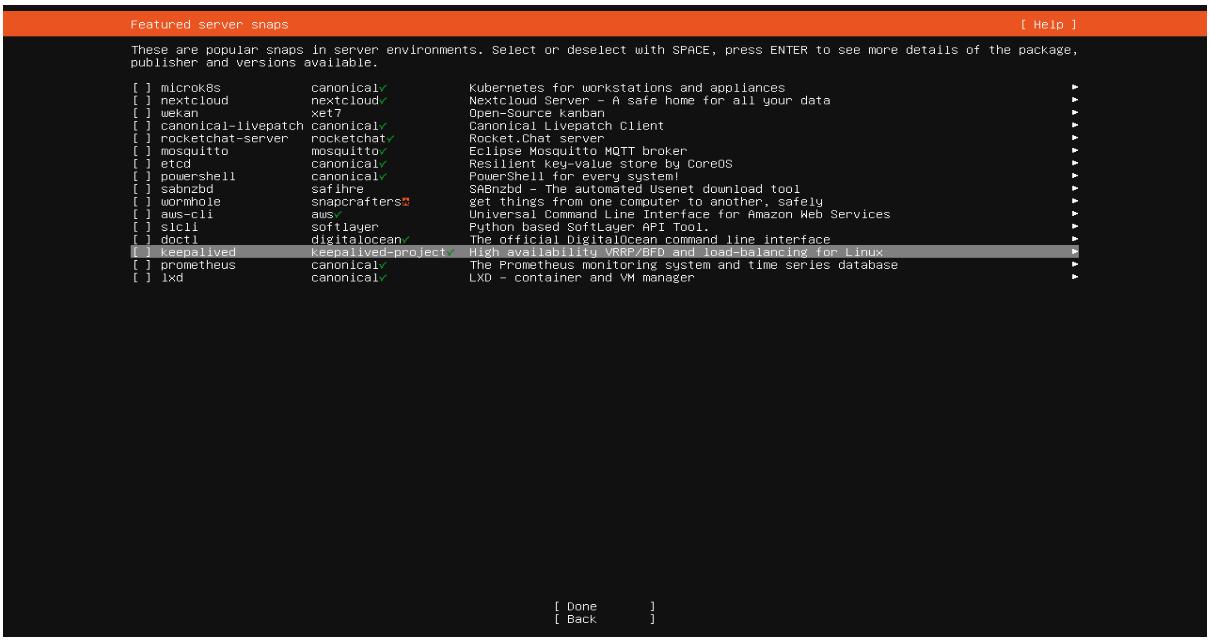
2.2.8 Additional Packages

The laboratory specifically says:

> Do not install additional packages unless instructed.

Therefore, skip additional packages.

Don't select extra applications just because they are available.



2.2.9 Complete Installation

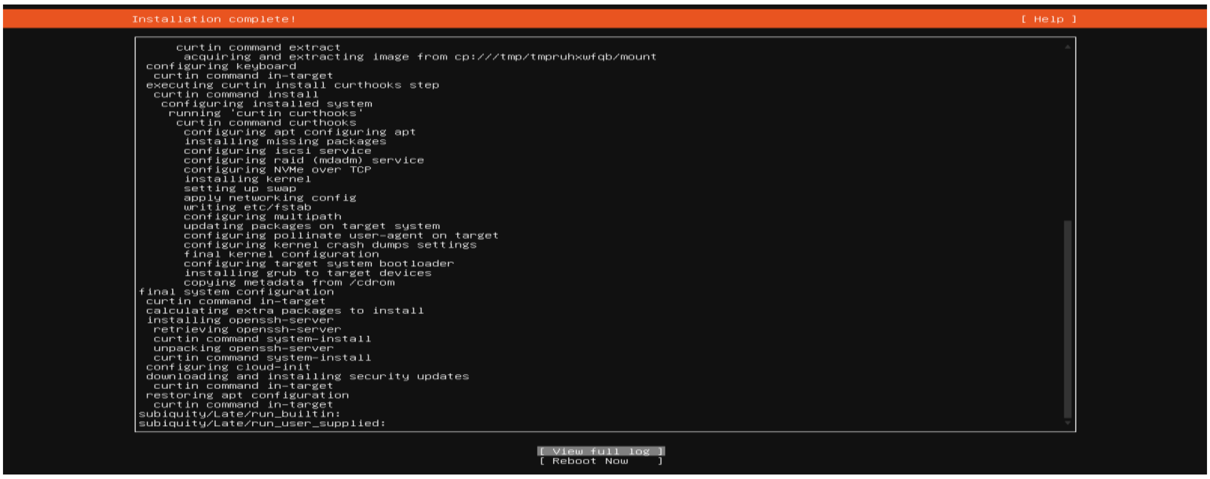
Allow Ubuntu Server to complete the installation.

Wait for the installation to finish.

When prompted, restart the virtual machine.

If Ubuntu tells you to remove the installation media:

1. Remove/detach the Ubuntu ISO from VirtualBox.
2. Restart the VM.



3. Initial Server Configuration

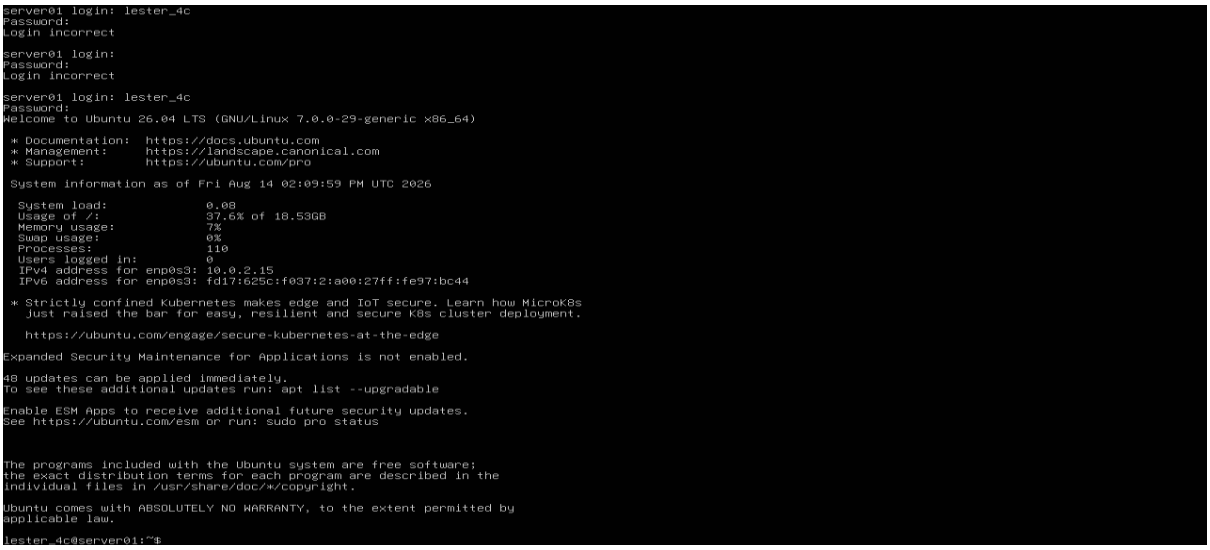
Configuration	Required Setting
Hostname	server01
Administrative User	Non-root user
Network	DHCP
SSH	OpenSSH Server installed
Storage	Guided – Use Entire Disk
Additional Packages	None

4. Server Verification Procedures

This section verifies that the Ubuntu Server was installed and configured correctly. The procedures confirm successful login, hostname configuration, IP address assignment, internet connectivity, system updates, and SSH service operation. Each verification task includes the required command and a screenshot of the result as evidence that the server is functioning properly

4.1 Login

After restarting the VM, log in using the account you created during installation.



4.2 Verify Hostname

Run: hostname

Expected result: server01

```
Login incorrect

server01 login:
Password:
Login incorrect

server01 login: lester_4c
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Aug 14 02:09:59 PM UTC 2026

  System load:             0.08
  Usage of /:              37.6% of 18.53GB
  Memory usage:            7%
  Swap usage:              0%
  Processes:               110
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe97:bc44

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

lester_4c@server01:~$ hostname
server01
lester_4c@server01:~$
```

4.3 Verify IP Address

Run: ip addr

Find the active network interface and its assigned IP address.

```
  System load:             0.08
  Usage of /:              37.6% of 18.53GB
  Memory usage:            7%
  Swap usage:              0%
  Processes:               110
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe97:bc44

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lester_4c@server01:~$ hostname
server01
lester_4c@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:97:bc:44 brd ff:ff:ff:ff:ff:ff
    altname enx08002797bc44
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86284sec preferred_lft 86284sec
    inet6 fd17:625c:f037:2:a00:27ff:fe97:bc44/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86285sec preferred_lft 14285sec
    inet6 fe80::a00:27ff:fe97:bc44/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
lester_4c@server01:~$
```
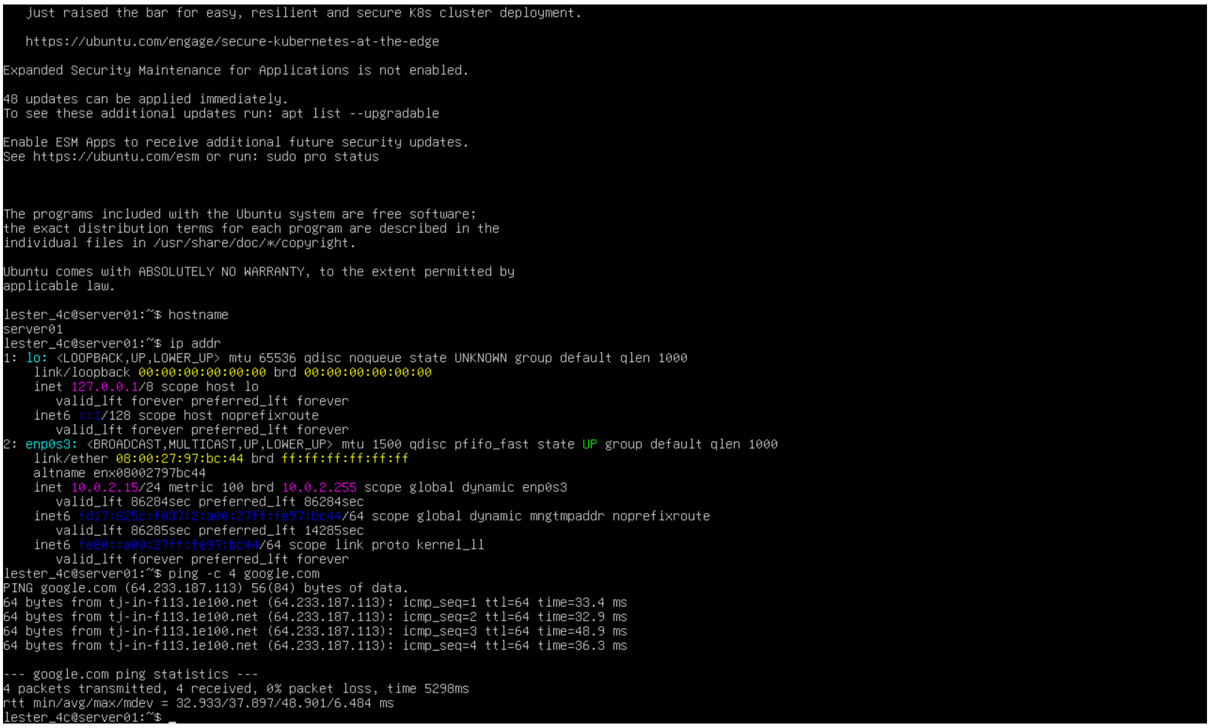

4.4 Test Internet Connectivity

Run exactly: ping -c 4 google.com



```
  just raised the bar for easy, resilient and secure K8s cluster deployment.

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lester_4c@server01:~$ hostname
server01
lester_4c@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:97:bc:44 brd ff:ff:ff:ff:ff:ff
    altname enx08002797bc44
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86284sec preferred_lft 86284sec
    inet6 fd17:625c:f037:2:a00:27ff:fe97:bc44/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86285sec preferred_lft 14285sec
    inet6 fe80::a00:27ff:fe97:bc44/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
lester_4c@server01:~$ ping -c 4 google.com
PING google.com (64.233.187.113) 56(84) bytes of data.
64 bytes from tj-in-f113.1e100.net (64.233.187.113): icmp_seq=1 ttl=64 time=33.4 ms
64 bytes from tj-in-f113.1e100.net (64.233.187.113): icmp_seq=2 ttl=64 time=32.9 ms
64 bytes from tj-in-f113.1e100.net (64.233.187.113): icmp_seq=3 ttl=64 time=48.9 ms
64 bytes from tj-in-f113.1e100.net (64.233.187.113): icmp_seq=4 ttl=64 time=36.3 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 5298ms
rtt min/avg/max/mdev = 32.933/37.897/48.901/6.484 ms
lester_4c@server01:~$ _
```
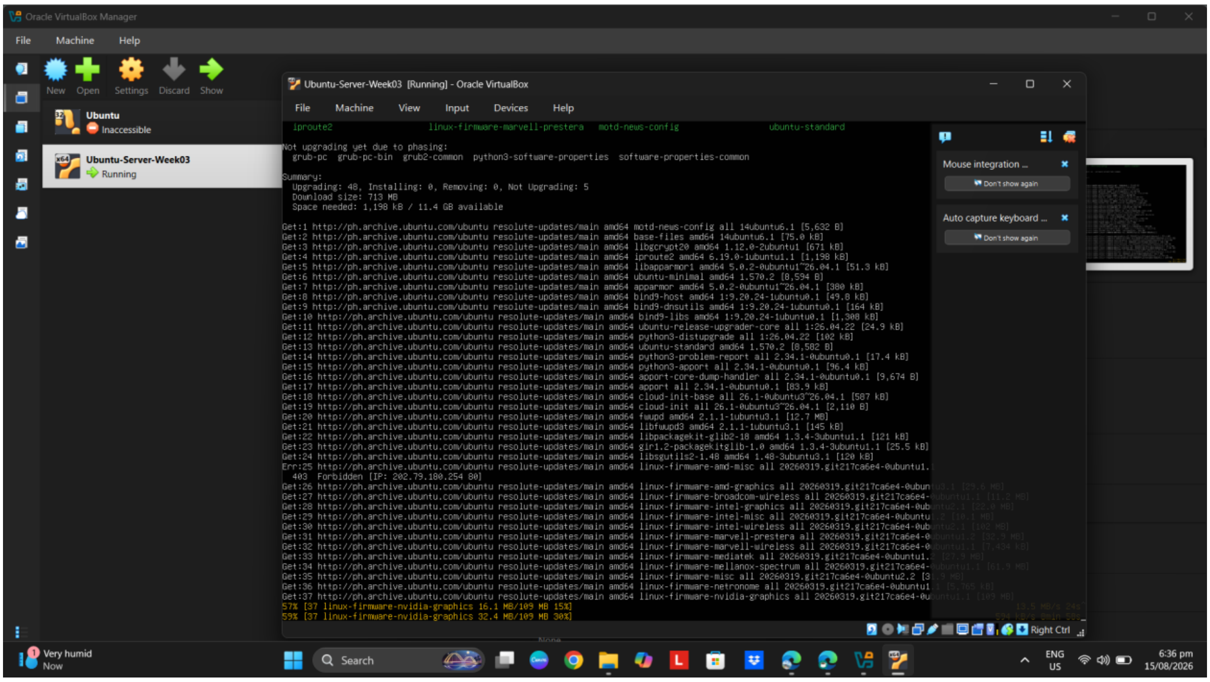

4.5 Update the Server

Run: sudo apt update

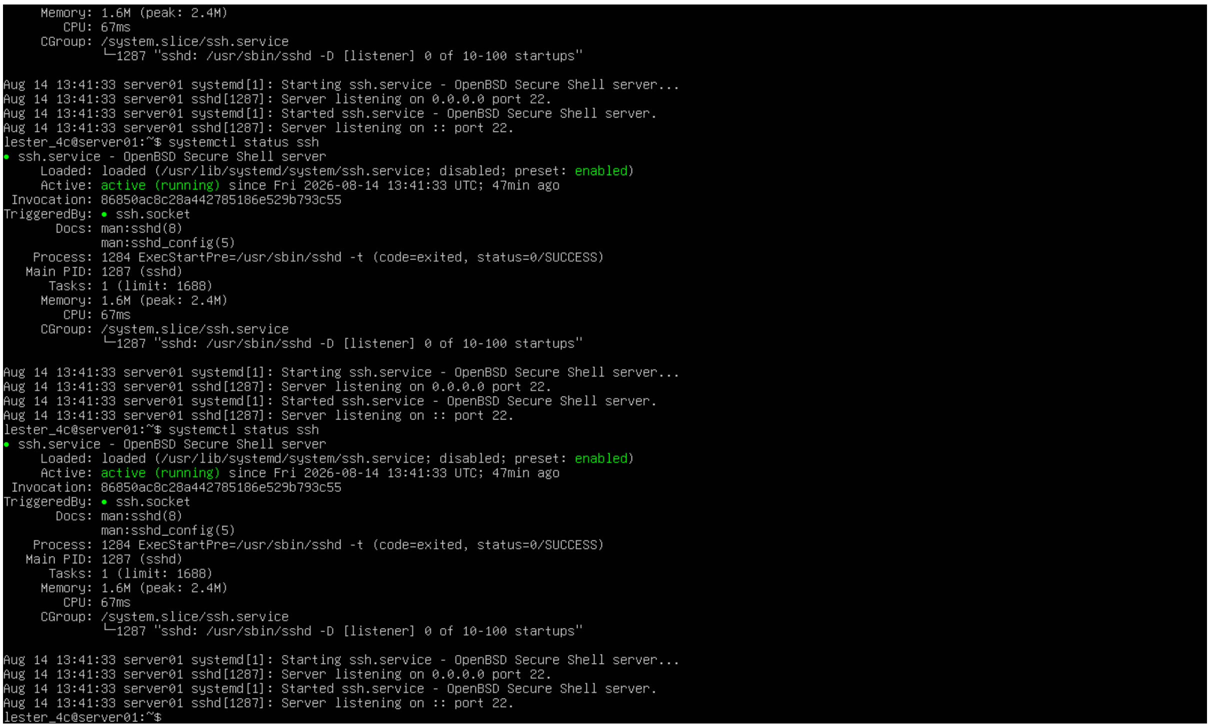
After that completes, run: sudo apt upgrade -y



4.6 Verify SSH Service

Run: systemctl status ssh

Look for: Active: active (running)



5. Troubleshooting Guide

Problem	Possible Cause	Solution
Ubuntu ISO takes a long time to download	Slow internet connection or large ISO file	Wait for the download to finish and verify the ISO before attaching it to VirtualBox
VM cannot boot	ISO not attached correctly	Check VirtualBox Storage settings
Internet ping fails	Network or DNS issue	Check ip addr and test connectivity
SSH is not running	OpenSSH was not installed or service stopped	Verify with systemctl status ssh
Wrong hostname	Incorrect hostname entered during setup	Check with hostname and correct the hostname if required
Installation is slow	Limited VM resources	Verify that the VM has at least 4 GB RAM and 2 processors

6. Best Practices

I learned that following best practices during Ubuntu Server installation and configuration is important to keep the server stable, secure, and easier to manage. I made sure that the virtual machine followed the required specifications, including 4 GB of RAM, 2 virtual processors, 40 GB of storage, and NAT networking. I also used a non-root administrative account for normal server administration instead of directly using the root account. After the installation, I updated the system using sudo apt update and sudo apt upgrade to make sure that the available packages were up to date. I also avoided installing unnecessary packages and enabled only the required service, which was OpenSSH. Lastly, I documented important server information such as the hostname, IP address, SSH status, and verification results. These practices helped me understand how proper server configuration and documentation can make system administration more organized and reliable.

7. References

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